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KSERC & RIET: AI Truing-Up Tool Project Overview

Aim: Presenting this prototype to KSERC to gather expert insights, ensuring our AI algorithms perfectly align with your strict regulatory vision for the 2024-25 audits.

> ***Project Insight:*** *An advanced AI regulatory assistant automating complex mathematical audits, designed specifically to ensure KSERC retains absolute control over all final verdicts.*

1. Human-in-the-Loop Model

- **Manual Edits:** Edit $Allowed_Amounts$ to overrule AI calculations instantly.
- **Override Reasoning:** Type custom justifications directly into the generated review tables.
- **Change Verdicts:** Set $Approval_Verdict$ (Approve/Partially Approve/Disallow) for any cost.
- **Modify Inputs:** Change $Target_Loss$ or $Base_Revenue$ to instantly recalculate the entire document.

2. Normative Formulas Engine

The AI does not guess numbers. It calculates data purely using deterministic KSERC Tariff Regulations algorithms:

- **O&M Calculation:** $O\&M_t = O\&M_{t-1} \times Index_{inflation}$
- **Revenue Gap:** $Gap = Rev_{actual} - ARR_{approved}$
- **Precedent Check:** $Claim_{current} \leftrightarrow DB_{historical_decisions}$

Core Formulas Simplified

Regulatory Cost Head	Algorithmic Formula	Description	Official's Benefit
O&M Escalation	$O\&M_{new} = O\&M_{base} \times (w_1 AICPI + w_2 WPI)$	Escalates base O&M using AICPI and WPI limits.	Prevents inflating employee/admin costs beyond inflation limits.
Interest on Surplus	$Int = S_{accum} \times R_{sbi}$	Multiplies accumulated surplus (S_{accum}) by SBI rate.	Automates the guaranteed consumer benefit from past surplus.
Depreciation Cap	$If(\Sigma Dep \geq 0.9 \times C_{asset}) \rightarrow Dep_{current} = 0$	Halts calculation at 90% of original asset cost.	Stops accidental over-recovery on fully depreciated assets.
Loan Interest	$Int = (C_{asset} \times 0.70) \times R_{norm}$	Caps interest using normative 70:30 Debt-Equity ratio.	Ensures consumers only pay for approved prudent borrowing.

3. Distributive Laws & Penalties (Version 6) In-Progress.

- **Loss Flagging:** Automatically triggers an alert when $Loss_{actual} > Loss_{target}$.
- **Structural Disallowance:** Algorithm computes financial penalty instantly: $Penalty = (Loss_{actual} - Loss_{target}) \times Cost$.
- **Consumer Protection:** Structurally guarantees that excess loss recovery is permanently and automatically disallowed.

4. Anticipated Queries & Expert Considerations

Given the years of truing-up experience of KSERC, we anticipate the following critical questions:

- **Data Accuracy:** *"How do we trust AI extraction?"* The system displays original PDF snippets directly next to AI data for immediate verification.
- **Rule Amendments:** *"What if KSERC updates tariff regulations?"* Our mathematical engine is modular. We simply update the configuration file, not the whole AI.
- **Unique Claims:** *"How are novel, unmapped expenses handled?"* AI flags unknown claims instantly, strictly routing them to officials for manual human judgment.
- **Audit Trail:** *"Can we track human overrides?"* Yes. The system logs every manual edit, preserving a permanent record of your regulatory decisions.